CLOUDIE: SECURE CLOUD STORAGE
WITH LAYERED MALWARE DETECTION & AES-256 ENCRYPTION

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INTRODUCTION

Cloud storage is where people keep their most important files, and where malware travels with them. A file that looks clean today may be **recognised as malware next week**, and files stored unencrypted are exposed the moment storage is breached. **Cloudie** scans every upload with several independent checks, encrypts it with **AES-256** before it is stored, and **keeps checking it after upload**.

PROBLEM STATEMENT

01 New and disguised malware slips past signature scanners

02 Files are scanned once: one found to be malware later stays "clean"

03 Stored files and logs can be read or quietly altered after a breach

OUR SOLUTIONS

LAYERED MALWARE ANALYZER
- **VirusTotal** (~70 antivirus engines) is the primary check
- **AI Random Forest** reads 32 structural features of PDFs
- **Structure checks** catch macros, DDE, zip slip, fake extensions

AES-256 ENCRYPTION
- **Unique key and IV** for every file
- Only a **SHA-256 hash** of the key is stored, never the key
- File hash re-checked after decryption (integrity)

QR CODE KEY
- Key shown once as **text, QR and key file**, and emailed
- Unlock by **scanning the QR** with the camera
- Wrong key? Cloudie names the file it belongs to

INNOVATION

RETROACTIVE RE-CHECK
- Asks VirusTotal **again after upload**, using only the file's fingerprint
- **No key needed**, the file is never decrypted
- Flagged later? File is **quarantined** and the user warned

TAMPER-EVIDENT AUDIT
- Every action sealed in an **HMAC-SHA256 hash chain**
- Edited or deleted entries show **"Tampering detected"**

SECURE ACCESS
- Email verification and **2-step login code**
- Rate limits stop password guessing
- The AI scanner is private, not on the internet

SOCIAL & ECONOMIC IMPACT

01 **LAYERED SECURITY**
Prevents (scan before storing), protects (AES-256) and keeps watching (re-check, audit).

02 **SOCIAL IMPACT**
Aligned with UN SDG 9 (Industry, Innovation & Infrastructure) and SDG 16 (Peace, Justice & Strong Institutions).

03 **COST-EFFECTIVE**
Open-source tools and Google Cloud services that scale to zero when nobody is using them.

04 **COMMUNITY FRIENDLY**
Runs in any browser or phone, nothing to install. The QR key works for non-technical users.

05 **PRIVACY BY DESIGN**
Encryption keys are never stored at rest, so a database or storage leak alone opens no file.

06 **HONEST & EXPLAINABLE**
Every verdict names its reason, and the limits are measured and published, not hidden.

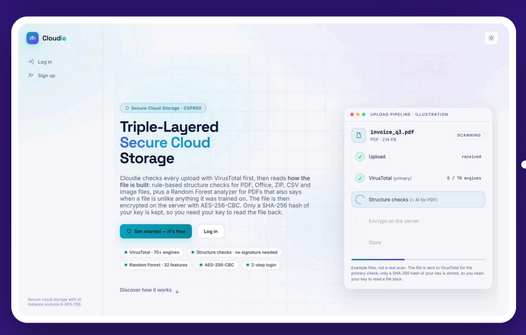
99.87% AI accuracy on held-out PDFs (18,126-file dataset)

1.0% of 200 real-world PDFs tagged, 0 blocked (false alarms)

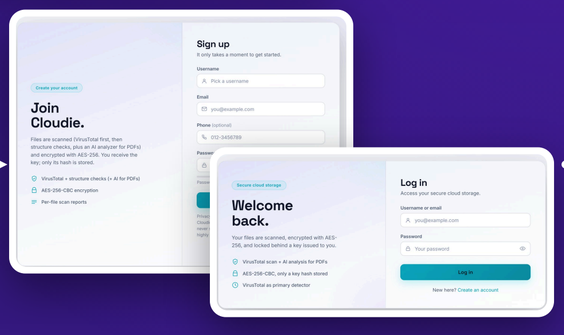
~70 antivirus engines check every upload

280 automated tests (92 PHP + 188 Python)

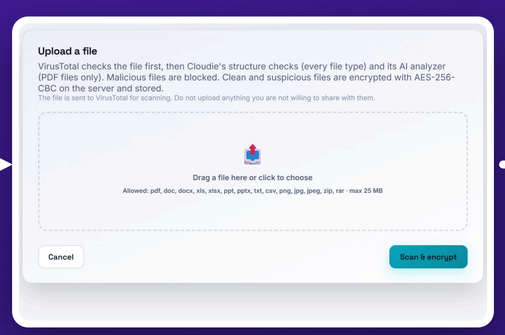
INVENTION APPLICABILITY



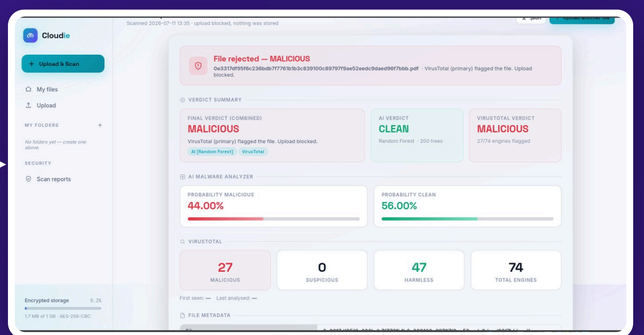
Homepage



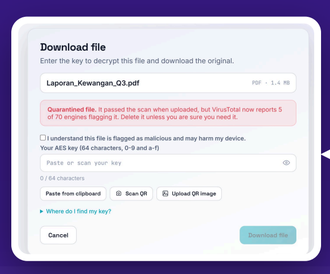
Sign up & 2-step login
password + emailed one-time code



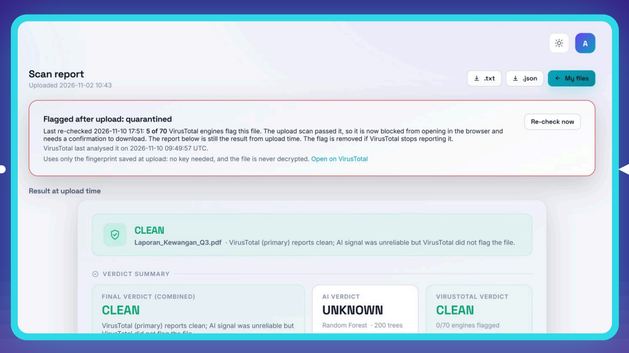
Upload & scan
VirusTotal, structure checks, AI



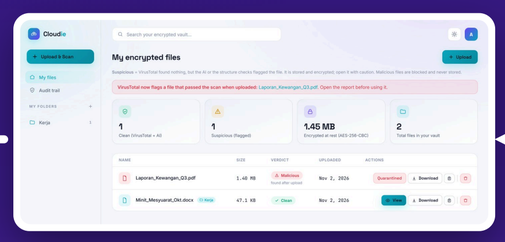
Malware: blocked, never stored



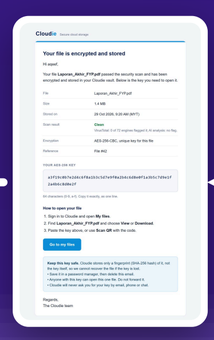
Decrypt with key or QR
quarantine needs a warning



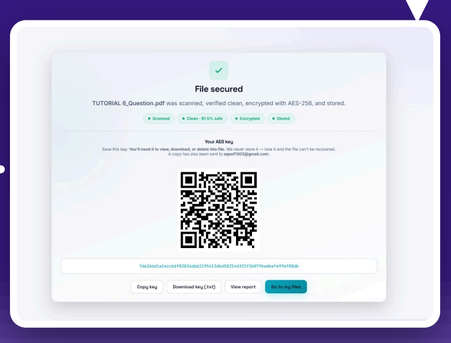
Retroactive re-check
VirusTotal flags it after upload: quarantined



My encrypted files
re-checked in the background



Key email

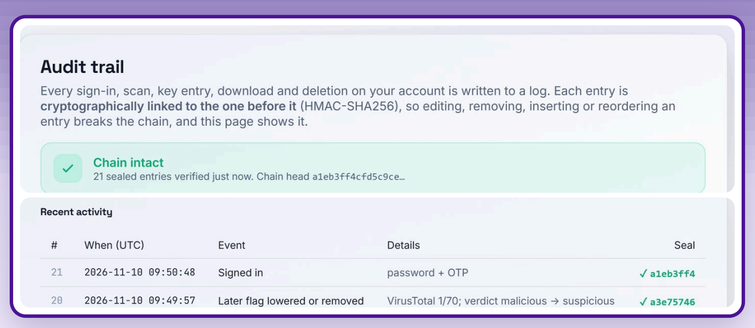


Clean: encrypted, key + QR
shown once, only its hash kept

TARGET USERS

- Individuals & students
- Small businesses (SMEs)
- Government sector
- NGOs
- Universities & schools

TAMPER-EVIDENT AUDIT



Every action is sealed in a hash chain. Edit one row in the database and the page shows **"Tampering detected"**.

MAPPED FRAMEWORKS

Controls are **mapped** (not certified) to recognised security frameworks:

MITRE ATT&CK · NIST CSF 2.0 · ISO/IEC 27001:2022 Annex A · OWASP Top 10 · NIST SP 800-63B

Try it live

Source code